

cobas e801

Adjustment Procedure

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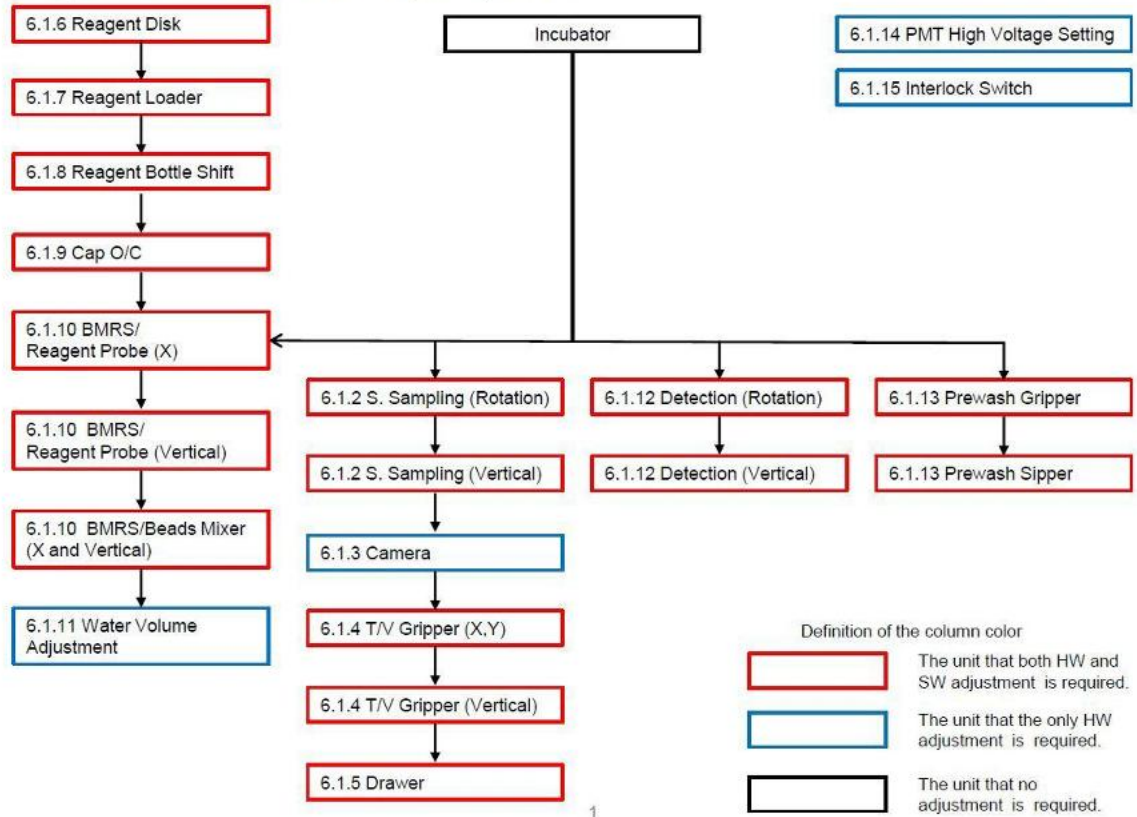
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1.0 Procedure of Adjustment

Ver. 01(June 2015)

6.1.1. Structure of e801 adjustment procedure

The following flow chart shows the structure of e801 adjustment procedure.



1

2.0 Reagent Disk

2.1 Jigs

Name	jig
Parts No.	788-9880
Name of item	Reagent Compartment Jig
Quantity(pos)	2
Adjustment position	Adjust the rotation direction of the R disk. Adjust the slope and the height of the beads mixing position.
Image or drawing	Use jigs with the sign on the bottom side. Upper side of jig (Type B) Front side of jig (Type B) jig (Type C)

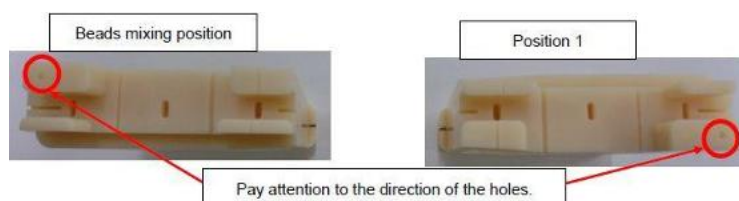
2.2 Preparations

Remove following Covers:

BMRS Cover, Gondola Cover, R Jacket Cover 1, R Jacket Cover 2, Loader Cover, Table Cover

2.3 Preparation Software Adjustment

- Start SSW
- Select **“Service – Adjustment - R.Pack Setting”** and click Adjust (Figure 2.1)
- Click **“Start”** (Figure 2.2)
- Rotate R. Disk by hand, place the Jig **Type B** in Position 1 and 48
- Adjust the positions of the Jigs **Type B** in the Bead Mixing Pos. and Pos. 1, so that the grooves of the Jigs match as shown in Figure 3.2 by using pulse adjustment buttons.

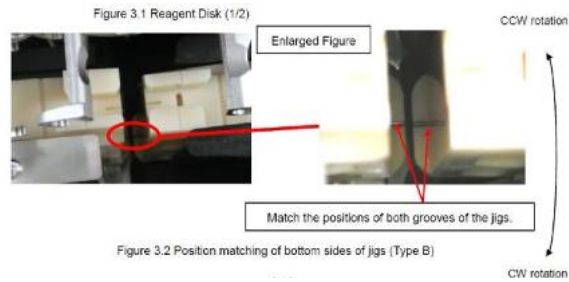
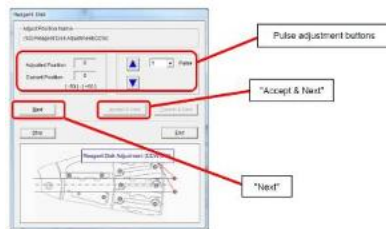


2.4 Software Adjustment

- Start SSW
- Select **“Service – Adjustment – Reagent Disk”**
- Before starting, be sure that the hardware adjustment finished successful and Jig **Type B** is placed.

Reagent Disk Adjustment (CCW)

- Click **“Accept”** and **“Next”**



Reagent Disk Adjustment (CW)

- Adjust the positions of the **Jigs Type B** in the Bead Mixing Pos. and Pos. 1, so that the grooves of the Jigs match as shown in Figure 3.2 by using pulse adjustment buttons.
- Click **“Accept”** and **“Next”**

Adjustment Check

- Check that the grooves on the bottom side of both **Jigs Type B** match in both Pos. (CCW and CW)
- Press the **“End”** button

Removing the Jigs Type B

- Select **“Service – Adjustment – R.Pack Setting”** and click **“Adjust”**
- Click **“Start”** and remove the Jigs by turning the rotor.
- Press the **“End”** button

3.0 Reagent Loader

3.1 List of Jigs

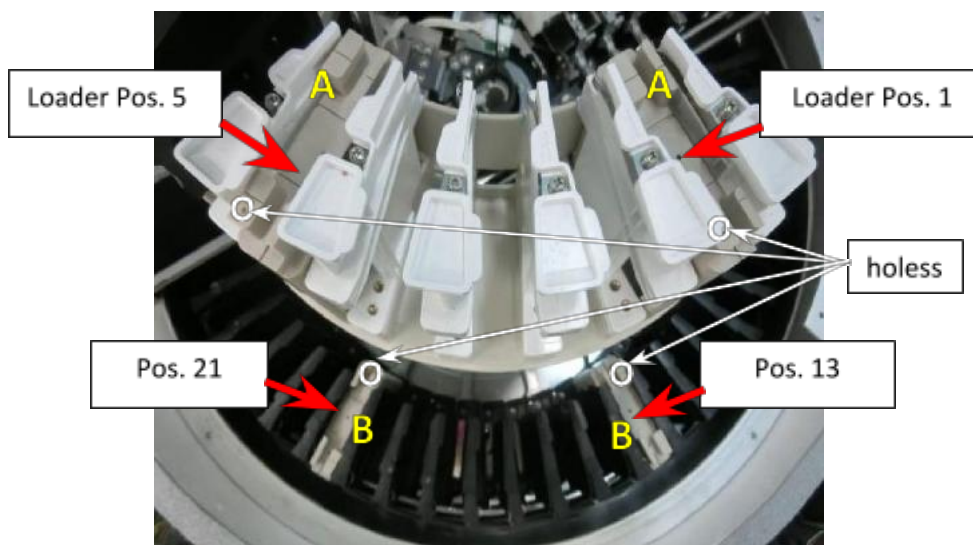
Name	jig
Parts No.	788-9880
Name of item	Reagent Compartment Jig
Quantity(pcs)	1
Adjustment position	Positioning this assembly and the R disk. Adjusting horizontally.
Image or drawing	

3.2 Preparation

remove following covers: BMRS cover, Gondola cover, R jacket cover 2 (RJ2 cover S), Loader cover, Table cover, R jacket cover 1

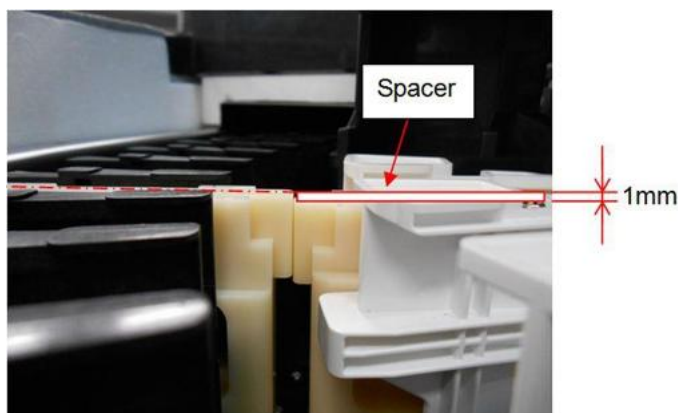
3.3 Preparation Software Adjustment

- Start up SSW, select "**Service - Adjustment - Reagent Loader**", and then click "**Adjust...**"
- After reset, click "**Next.**"
- After the bottle base has ascended, place the **Jig Type A** in the position (Loader Pos. 1 and 5) shown in Figure below and the **Jig Type B** in the corresponding position of R disk (Pos. 13 and 21). Place each of the jigs in the loader and R disk positions paying attention in order not to place them in a wrong direction (check holes on jig) and pushing them firmly into the slots.



3.4 Adjustment of height and horizontal position to R disk

- When the jigs are placed properly press “**Next**” in the SSW.
- After the bottle base has descended , place four Jig C on top of two Jig A and two Jig B.
- Move the Loader up and down by using Pulse adjustment button in SSW , until the height of **Pos.5** and **R Disk jigs** (Type C) are aligned .
- Put 1 mm spacer on the Pos.1’s jig and consider R Disk side’s jig is higher

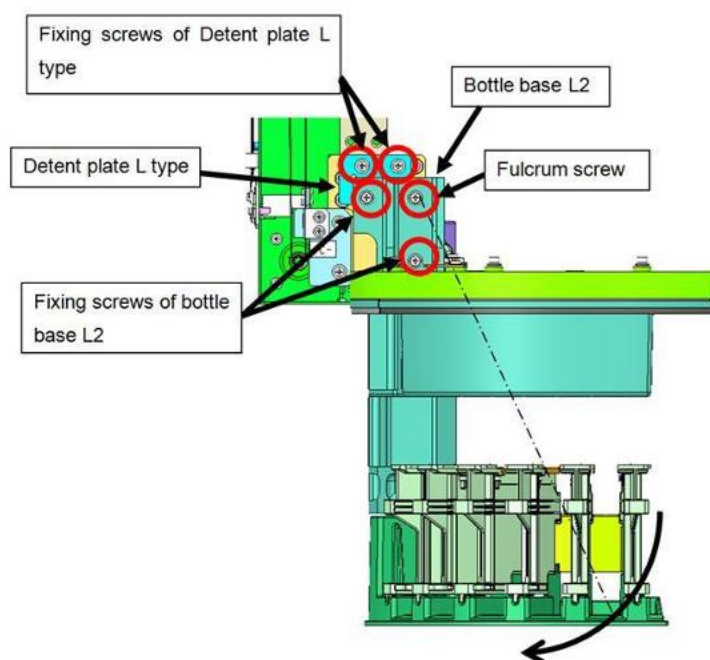


- If the gap Pos.1 against Pos.5 is not 0-1.0mm , adjust the slope by the following procedure ;

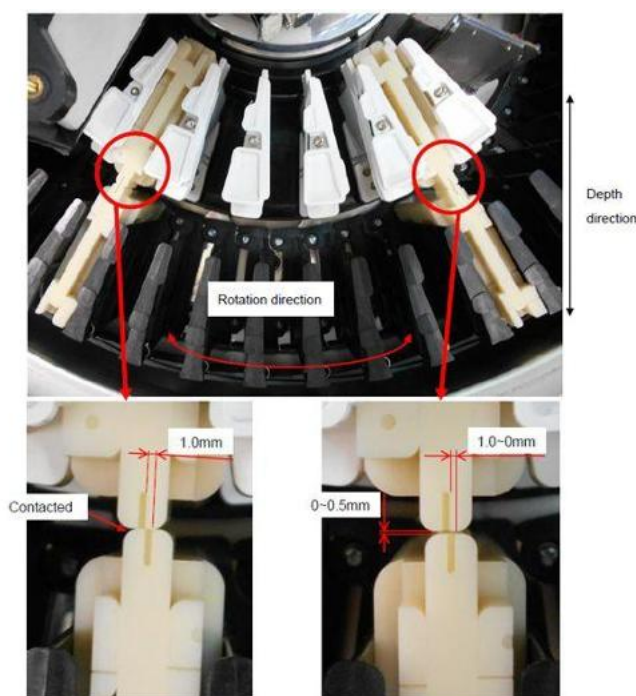
Loosen the Fixing screws of the Detent plate L type and Fixing screws of bottle base L2.

As to Fixing screws of bottle base L2 , make sure to loosen only the adjustment 2 screws.

- When you rotate it to the direction as shown in the figure below , Pos.1 will go down.
- When you rotate it reverse , Pos.1 will go up.



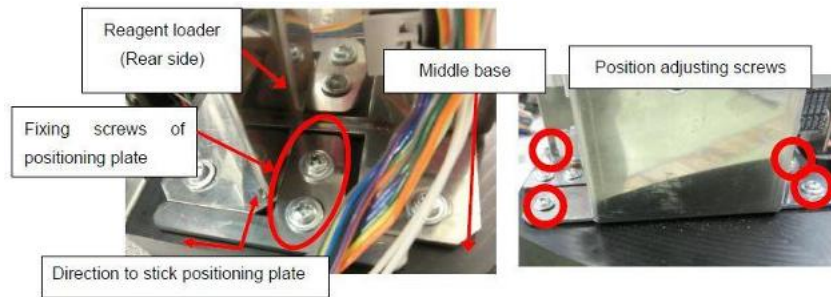
- Rotate the Bottle base L2 on a central point, the Fulcrum screw; to make the height of POS1 to POS5 0~ -1.0mm (0mm is the best).
- After the adjustment, fix the Bottle base L2, adhere the Detent plate L type to the Bottle base L2 and fix the plate.
- Move the Loader up and down by using Pulse adjustment button in SSW, until the height of POS5 and surface of R disk Jig (Type C) are aligned.
- After that, lower the loader for 7 pulses by using Pulse adjustment button in SSW.
- Pull the detent plate upward to make the height of POS1 to surface of R disk jig (Type C) for -0.5~0mm (-0.5mm is the best). After the alignment, fix the detent plate at that position.
- After adjusting for the height , check the positioning of the Reagent Loader against Reagent Disk as below ;



Caution : Make sure that jigs (Type C) does not contact the other jig hardily.

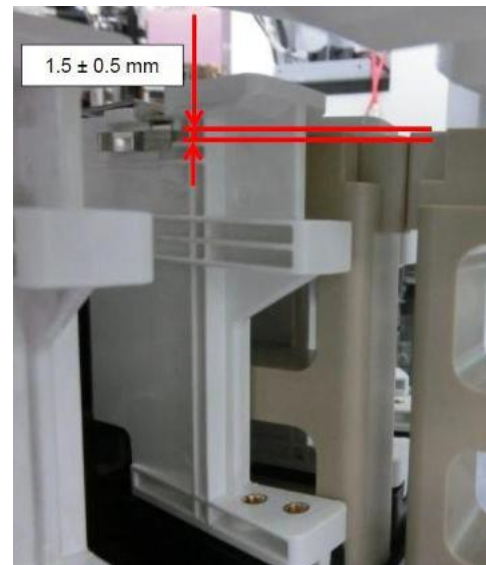
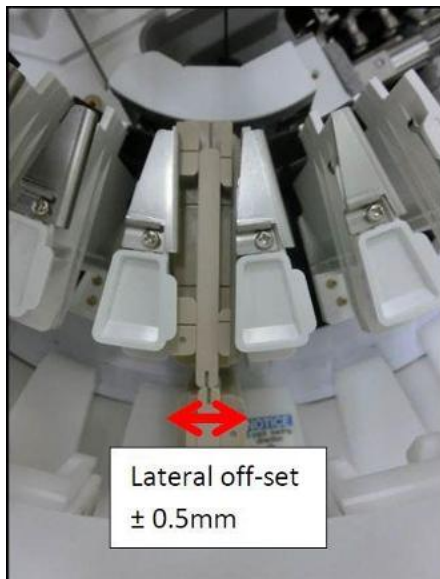
- Use the four adjustment screws of Reagent Loader for this adjustment
- Arrange the Reagent Loader in the position that satisfies following conditions
 - Groove of Pos.5 Jig is off-center to the left about 1.0 mm against R Disk Jig
 - Each jig is in contact.
- At this time , confirm that groove of Pos.1 Jig is off-center to the left 0~1.0 mm and gap between each Jig is 0~0.5 mm .Use Jig as a reference because its groove is just 1.0 mm.

- After the adjustment , adhere the Positioning plate to the Reagent Loader and fix the plate.



3.5 Adjustment of height and horizontal position for loading

- Hint: this adjustment can only be performed when the Pre-opener is attached to the instrument
- Start up SSW, select "**Service - Adjustment - Reagent Loader**", and then click "**Adjust...**"
 - After reset, click "**Next.**"
 - After the bottle base has ascended, place the **Jig Type A** in the loader position 3, and the **Jig Type B** in the Pre-opener Pos. 3. Place the **Jigs Type C** onto the **Jigs A and B** (see figure below)
 - The lateral off-set should be less than 0.5mm
 - The jig on the loader side should be lower than on the Pre-opener side. The height difference should be $1.5 \pm 0.5\text{mm}$
 - To change the height difference move the loader up or down stepwise with the SSW



- Remove all the jigs by hand
- After adjustment press "**Next**" until the Loader descends, then press "**End**" and close the SSW
- Attach following covers: BMRS cover, Gondola cover, R jacket cover 2 (RJ2 cover S), Loader cover, Table cover, R jacket cover 1

4.0 Reagent Bottle Shift

4.1 List of Jigs

Name	R. pack	Jig
Parts No.	-	788-8880
Name of item	R. pack	Reagent Compartment Jig
Quantity(pcs)	5	2x for Jig (Type A), 1x for Jig (Type B), 1x for Jig (Type C)
Adjustment position	RS position, BM position, and Loader position	
Image or drawing		

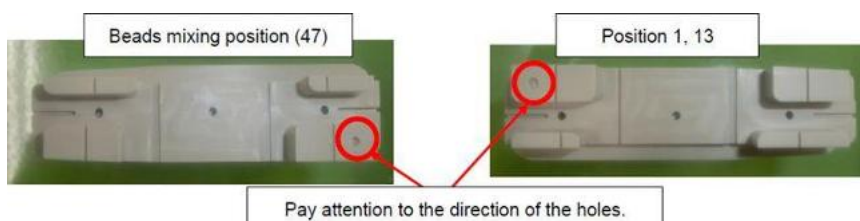
4.2 Preparation

Remove following Covers:

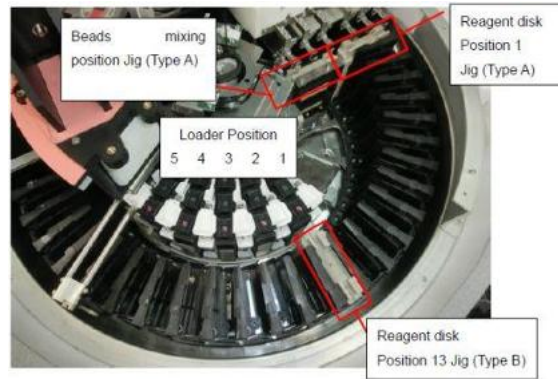
BMRS Cover, Gondola Cover, R Jacket Cover 1, R Jacket Cover 2, R Jacket Cover 3, Loader Cover, Table Cover, Top Cover (Not remove, just pull it upwards), Rear Cover

4.3 Preparations before Adjustments

- Start SSW
- Select **"Service – Adjustment – R.Pack Setting"**; then click **"Adjust"** and **"Start"**
- After reset, place the **Jig Type A** in Pos. 1 and 47
- Move the **Jig Type A** in Pos. 47 to the Bead Mixing Pos.
- Place a **Jig Type B** in Pos. 13
- Load 5 Reagent cassettes in the Reagent Loader Pos. 1-5
- Make sure to place it in the right direction and push it tightly.
- After Loading click **"End"**



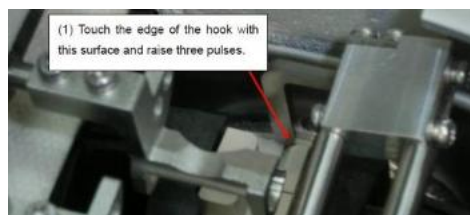
- Select **“Service – Adjustment – Reagent Bottle Shift”**; then click **“Adjust”**



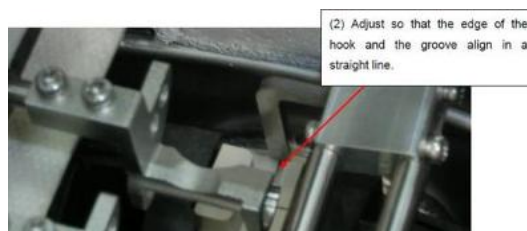
4.4 Software Adjustment

(1/11) Bottle Shift Arm Home Position (Rotate) (CW)

- Perform Pulse Adjustment that the Hook of the Bottle Shifter touches the surface of **Jig Type A**
- Raise it 3 pulses from the contact position.



- Perform the pulse adjustment of the arm in rotational direction so that the edge of the bottle shift hook and the groove of the **Jig Type A** align in a straight line.



- Click **“Accept”** and **“Next”**

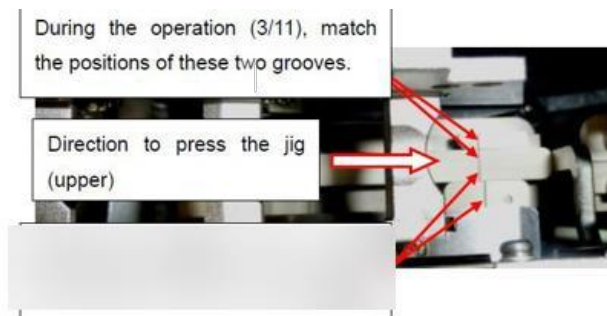
(2/11) Bottle Shift Arm Reagent Loading Position (Rotate) (CW)

- Check that the bottle shift arm rotates to the reagent loading position 13.
- Perform the pulse adjustment of the arm rotation direction so that the edge of the bottle shift hook and the groove of the **Jig Type B** align in a straight line.
- Click **“Accept”** and **“Next”**

(3/11) Bottle Shift Arm Home Position (X-Direction)

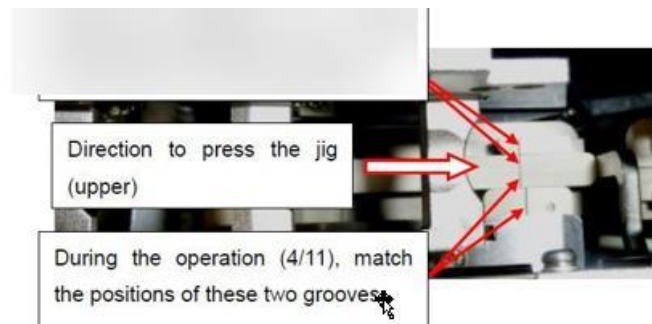
- Make sure that the **Jig Type A** is pushed into the deepest side of the inner Ring.
- Place the **Jig Type C** on the upper side of the **Jig Type A**

- Perform the pulse adjustment so that the positions of the grooves of the **Jig Type A** and the **Jig Type C** match.
- When checking the position, press the **Jig Type C** to the hook.
- Click **“Accept”** and **“Next”**



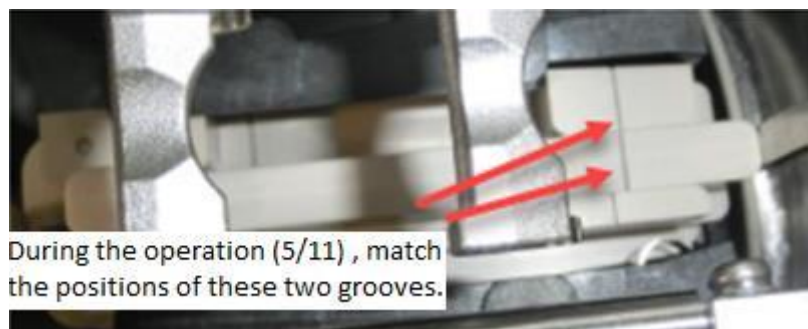
(4/11) Clearance of the hook and the R. Pack at inner position

- Check that the bottle shift hook is cleared from the **Jig Type C**.
- Perform the pulse adjustment that the positions of the grooves of the **Jig Type A** and the **Jig Type C** match.
- When checking the position, press the **Jig Type C** to the hook.
- **Caution:** Remove the **Jig Type C** before choosing the Accept & Next button.
- Click **“Accept”** and **“Next”**



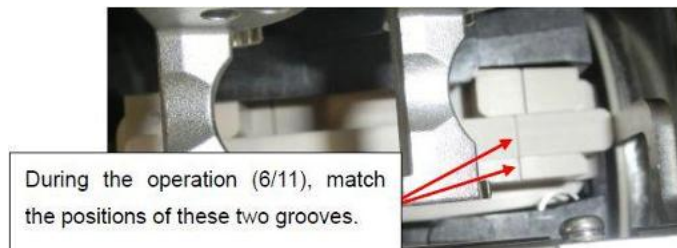
(5/11) Contact Position for the hook and the R. Pack at outer position

- Place the **Jig Type C** on the upper side of the **Jig Type A** on the side of the outer Ring.
- Perform the pulse adjustment so that the positions of the grooves of the **Jig Type A** and the **Jig Type C** match.
- When checking the position, press the **Jig Type C** to the hook.
- Click **“Accept”** and **“Next”**



(6/11) Clearance of the hook and the R. Pack at outer position (CCW)

- Check that the bottle shift hook is cleared from the **Jig Type C**.
- Perform the pulse adjustment that the position of the groove on the **Jig Type C** matches the groove on the front side of the **Jig Type A**.
- When checking the position, press the **Jig Type C** to the hook.
- **Caution:** Remove the **Jig Type C** before choosing the Accept & Next button.
- Click **"Accept"** and **"Next"**



(7/11) Bottle Shift Arm Reagent Loading Position 1 (Hardware adjustment)

- The hook moved to the center position of the bottle.
- Check that the hook does not touch the inner walls of the bottle shift and the reagent loader when the hook is moved up and down.
- Click **"Next"**



(8/11) to (11/11) Bottle Shift Arm Reagent Loading Position 2 ~ 5 (Hardware adjustment)

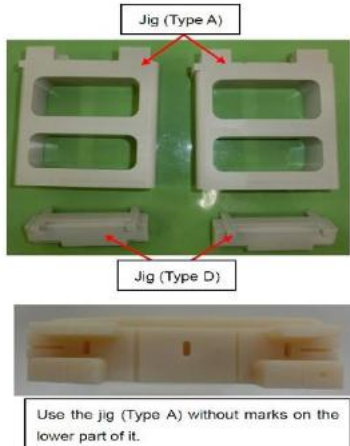
- Check the reagent loading position 54 ~ 57 in the similar way to (7/11) Bottle Shift Arm Reagent Loading Position 1 (Hardware adjustment).
- Perform (1/11) Bottle Shift Arm Home Position (Rotate) (CW).
- The hook does not touch the inner walls of the bottle shift or the reagent loader during the operation.
- Click **"End"**

Removing the Jigs

- Click **"End"**
- Select **"Service – Adjustment - R.Pack Setting"** and click **"Adjust"**.
- Click **"Start"**
- Remove the all Jigs by hand.
- Click **"End"**

5.0 CAPOC Adjustment

5.1 List of Jigs

Name	jig
Parts No.	788-9880
Name of item	Reagent Compartment Jig
Quantity(pcs)	1
Adjustment position	
Image or drawing	

5.2 Preparations

Remove following Covers:

BMRS Cover, Gondola Cover, R Jacket Cover 2, R Jacket Cover 3, Top Cover (not remove, pull it upwards), Rear Cover

5.3 Preparations Software Adjustments

- Start SSW; Select **“Service – Adjustment – R.Pack Setting”**; then click **“Adjust”** and **“Start”**
- After reset, place the **Jig Type A** in Pos. 2 and 48 (Refer Figure 2.3)
- Move the **Jig Type A** in Pos. 48 to the Bead MixingPos.
- Make sure to place it in the right direction and push it tightly.
- Click **“End”**

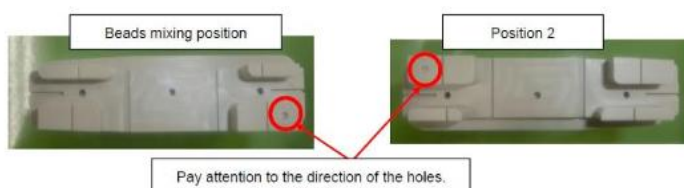
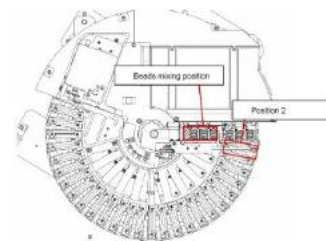


Figure 2.3 Direction to place jig



5.4 Software Adjustments

- Start SSW; Select **“Service – Adjustment – Reagent Cap O/C”** (Figure 3.1 displayed)
- Be sure that the **Jigs Type A** are in Bead Mixing Pos. and Pos. 2
- Place the **Jigs Type D** on both Jigs Type A
- Pay attention to the right direction and push them in without a gap. (Figure 3.3)

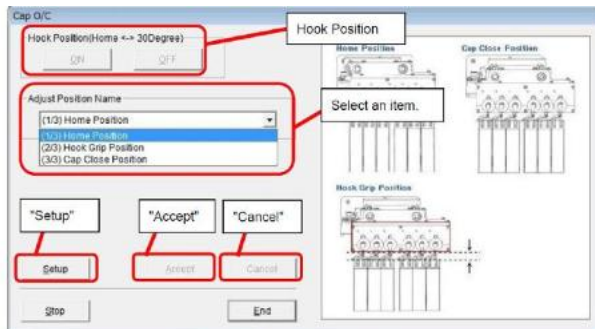


Figure 3.1 Adjustment screen

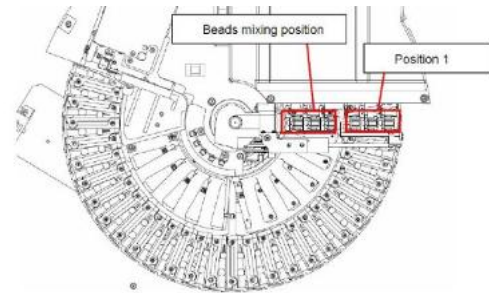


Figure 3.2 Position of jigs after resetting



Figure 3.3 Direction of placing jigs

Adjustment of CAPOC Position

- Set **“Hook Position”** to **“On”** and then to **“Off” (Important!)**
- Select **“(2/3) Hook Grip Position”** and Click **“Setup”**

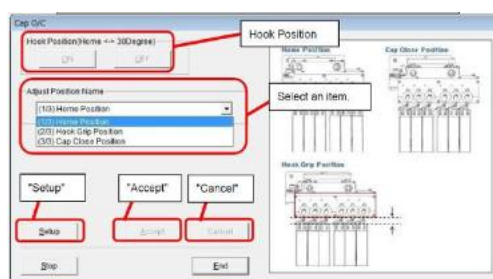
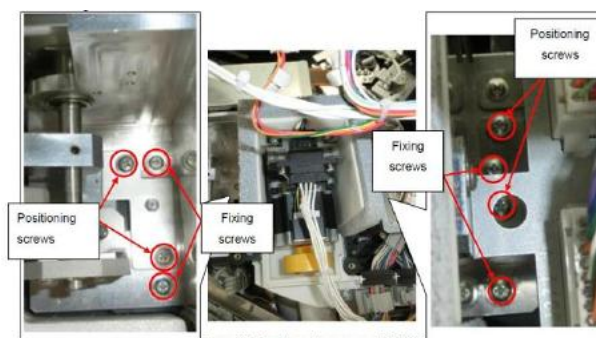


Figure 3.4 Adjustment screen

- CAPOC moves to **“Hook Grip Position”**
- For mech. adjustment loosen the four (4) fixing screws, **do not loosen completely!!** (Figure 3.5)

Figure 3.5 Positions of screws on CAPOC
10 / 13

- Move CAPOC towards the left side (Figure 3.6)
- Adjust the CAPOC that the convex part of Jig **Type D** is parallel to the hook.
- If it's not possible, move the CAPOC towards the back side, arrange parallel, loosen the screws of the hook, pull hook forward and perform a fine adjustment.
- After adjustment, tighten the four (4) fixing screws.
- Click **"On"** of **"Hook Position"**

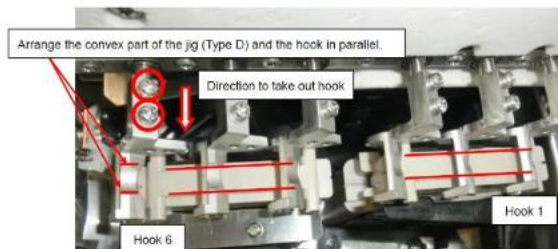


Figure 3.6 Arranging the CAPOC

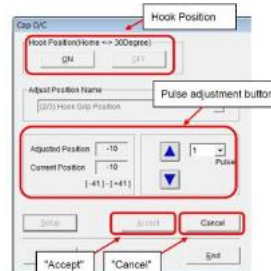


Figure 3.7 (2/3) Hook Grip Position

Hook Position Adjustment

- Using the Hook 6 as reference position, adjust X and Z direction according Figure 3.8

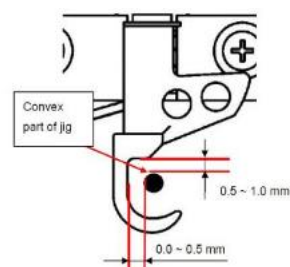


Figure 3.8 Gap between hook and jig (front side)

- Check the gap between Hook 6 and the **Jig Type D**. It should be in X-direction 0.0 ~ 0.5 mm and in Z-direction 0.5 ~ 1.0 mm. If the gap is within this range then click **"Accept"**
- Check the gap in X- and Z-direction between the hook and the **Jig Type D** also on hook 1-5. The gap should be within 0 ~ 1.5 mm. If the gap is within this range and then click **"Accept"**
- To correct the adjustment either use the SSW and move the CAPOC unit up and down stepwise or use the four (4) fixing screws and four (4) positioning screws (Figure 3.5) to move the CAPOC unit right or left.
- Set the **"Hook Position"** to **"On"**, set it to **"Off"** and check that the **Jig Type A** doesn't move.
- Set the **"Hook Position"** to **"Off"** and select **"(1/3) Home Position"** and click **"Setup"**
- Remove the **Jigs Type D**, then click **"Done"** and **"End"**
- If the **Jig Type D** is removed, click **"Yes"** (Figure 3.9)



Figure 3.9 Gap between hook and jig (front side)

Removing the Jigs **Type A**

- Start SSW; Select **"Service – Adjustment – R.Pack Setting"**; then click **"Adjust"** and **"Start"**
- Remove the **Jig Type A** and click **"End"**

Perform the maintenance function **Cap Opener Check** to confirm proper operation.

6.0 BMRS

6.1 List of Jigs

Name	Jig (1)	Jig (2)	Jig (3)
Parts No.	788-9880	-	788-3304
Name of item	Reagent Compartment Jig	AssayCup	BMRS jig
Quantity(pcs)	1	1	1
Adjustment position	Horizontal alignment of the reagent aspiration position Horizontal alignment of the beads mixing position	Horizontal alignment of the incubator position	Height alignment of the reagent aspiration position Height alignment of the beads mixing position
Image or drawing	 2x Jigs (Type A) 1x Jig (Type D) Use the jig (Type A) without a mark on the bottom part of it. 	  BM side (thick) RS side (thin) 	

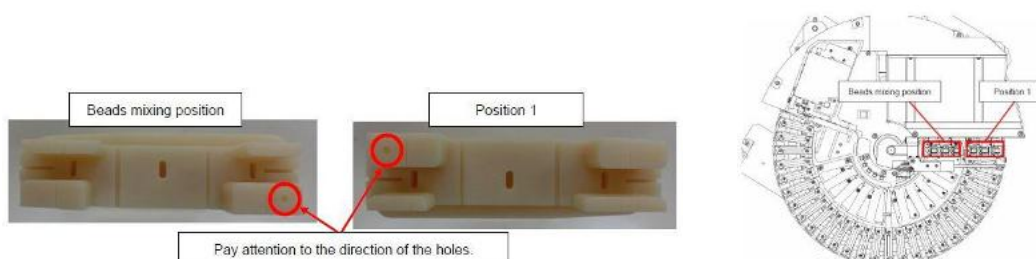
6.2 Preparations

Remove following Covers:

Gondola Cover, R Jacket Cover 2, R Jacket Cover 4, Loader Cover

6.3 Preparations Software Adjustments

- Start SSW; Select **“Service – Adjustment – R.Pack Setting”**; then click **“Adjust”** and **“Start”**
- After reset, place the Jig Type A in Pos. 1 and 47 (Refer Figure below)
- Move the Jig **Type A** in Pos. 47 to the Bead Mixing Pos.



- Make sure to place it in the right direction and push it tightly.
- Click **“End”**

6.4 Software Adjustment

Reagent Probe (X)

(1/9) Home Position (x)

- Place **Jig (Assay Cup)** in Position 48 of the Incubator.

- Pull the R2 Probe down.
- Perform Pulse Adjustment that the Probe is in the Center of the Jig. (Figures 3.1 and 3.2)
- Click **"Accept"** and **"Next"**

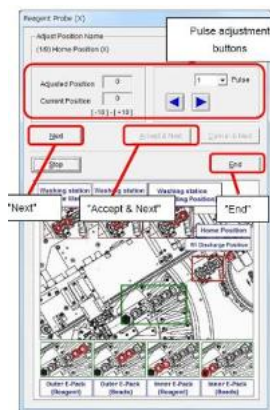


Figure 3.1 Initial screen of "Reagent Probe (X)"



Figure 3.2 Adjustment position of R2 probe

(2/9) R1 Discharge Position

- Pull the R1 Probe down.
- Perform Pulse Adjustment that the Probe is in the Center of the Jig. (Figure 3.3)
- Click **"Accept"** and **"Next"** and remove the Jig.



Figure 3.3 Adjustment position of R1 probe

(3/9) Outer R. pack (Reagent)

- Move both RS Probes down to **Jig Type A**. (Figure 3.4)
- Perform Pulse Adjustment to the Position so that each Probe is matched to the center of the hole of the Jig.
- Click **"Accept"** and **"Next"**

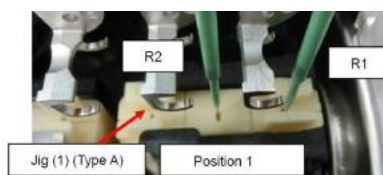


Figure 3.4 Outer R. pack (Reagent) adjustment position

(4/9) Outer R. pack (Beads)

- Move both RS Probes down to **Jig Type A**. (Figure 3.5)

- Perform Pulse Adjustment to the Position so that each Probe is matched to the center of the hole of the Jig.
- Click **"Accept"** and **"Next"**

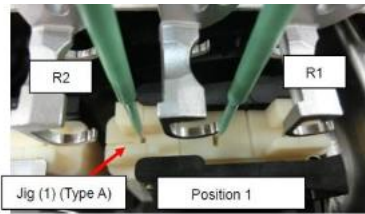


Figure 3.5 Outer R. pack (Reagent) adjustment position

(5/9) Inner R. pack (Reagent)

- Move both RS Probes down to **Jig Type A**. (Figure 3.6)
- Perform Pulse Adjustment to the Position so that each Probe is matched to the center of the hole of the Jig.
- Click **"Accept"** and **"Next"**

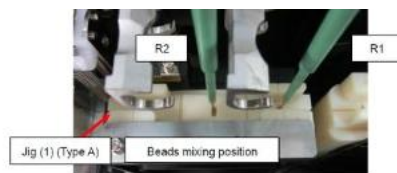


Figure 3.6 Inner R. pack (Reagent) Adjustment position

(6/9) Inner R. pack (Beads)

- Move both RS Probes down to **Jig Type A**. (Figure 3.6)
- Perform Pulse Adjustment to the Position so that each Probe is matched to the center of the hole of the Jig.
- Click **"Accept"** and **"Next"**

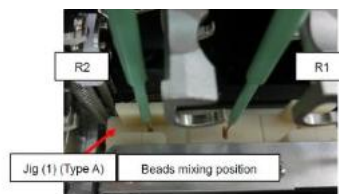


Figure 3.7 Inner R. pack (Beads) adjustment position

(7/9) Washing Station (Discarding Position)

- Perform Pulse Adjustment that the distance between Washing Station and R2 Probe is 33 ± 0.5 mm (Figure 3.8 and 3.9)
- Click **"Accept"** and **"Next"**

(8/9) Washing Station (CC Aspiration)

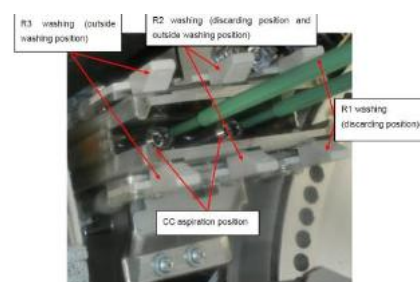


Figure 3.8 Washing position

- Perform Pulse Adjustment so that the distance between Washing Station and R2 Probe is 20 ± 0.5 mm (Figure 3.8 and 3.9)
- Click **"Accept"** and **"Next"**

(9/9) Washing Station (Outside Wash)

- Perform Pulse Adjustment so that the distance between Washing Station and R2 Probe is 7 ± 0.5 mm (Figure 3.8 and 3.9)
- Click **"Accept"** and **"Next"** and **"End"**

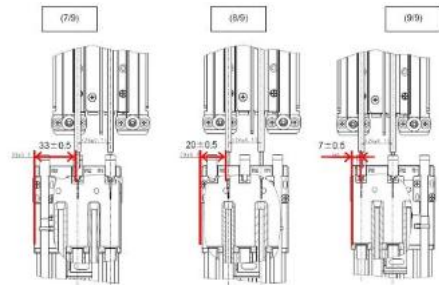


Figure 3.9 (7/9) ~ (9/9) adjustment position

Beads Mixer

Home Position (X)

- Select in SSW **"Service – Adjustment – BeadsMixer"**
- Select at Position name to adjust **"(1/3) Home Position (X)"**
- Click **"Setup"**
- Perform Pulse Adjustment so that the paddle is in the middle of the BM Washing Station. (Figure 3.11). The paddle must not touch the inner walls of the wash station.
- Click **"Accept"**

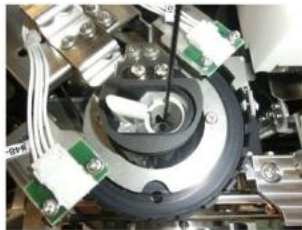


Figure 3.11 Home Position (X)

Beads Mixing Position (X)

- Select **"(2/3) Beads Mixing Position"**
- Place **Jig Type D** on **Jig Type A**
- Click **"Setup"**
- Perform Pulse Adjustment so that the paddle is located in the center of the hole. (Figure 3.12)
- Click **"Accept"**
- Remove the **Jig Type D**.

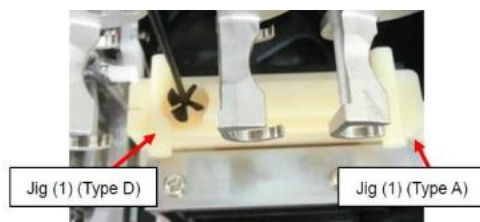


Figure 3.12 Beads Mixing Position (X)

Removing the Jigs Type A

- Click **"End"**

- Select **"Service – Adjustment - R.Pack Setting"** and click **"Adjust"**.
- Click **"Start"**
- Remove the **Jigs Type A** from BM Position and Position 1 by hand.
- Click **"End"**

Reagent Probe (Vertical)

(1/6) R1 Discharge Position (Vertical)

- Select **"Service – Adjustment – Reagent Probe (Vertical)"** and click **"Setup"**
- Choose Adjustment position name **"(1/6) Discharge Position (Vertical)"**
- Click **"Setup"** and **"Adjust"**
- Perform Pulse adjustment so that the R1 Probe touches the incubator. (Figure 3.14)
- Click **"Accept"**



Figure 3.14 Height over Incubator

(2/6) R2 Discharge Position (Vertical)

- Choose Adjustment position name **"(2/6) Discharge Position (Vertical)"**
- Click **"Setup"** and **"Adjust"**
- Perform Pulse adjustment so that the R2 Probe touches the incubator. (Figure 3.14)
- Click **"Accept"**

(3/6) R2 Aspiration Positions (Beads) (Vertical)

- Place the **Jig (3)** in the Beads Mixing Position that the thin side is located to the BM side. (Figure 3.15)
- Choose Adjustment position name **"(3/6) R2 Aspiration Positions (Beads)(Vertical)"**
- Click **"Setup"**
- The R Probe moves to the Beads Mixing Position.
- Move the **Jig (3)** to the Position right under the R2 Probe.
- Select **"Adjust"**
- Perform Pulse adjustment so that the R2 Probe touches the **Jig (3)**
- Click **"Accept"**



Figure 3.15 Height over the beads mixing position

(4/6) R1 Aspiration Positions (Reagent) (Vertical)

- Place the **Jig (3)** in the Position that the thin side is located to the R1 Probe side.

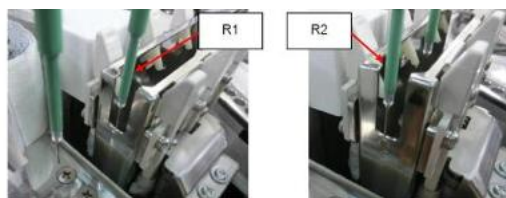
- Choose Adjustment position name **“(4/6) R1 Aspiration Positions (Reagent) (Vertical)”**
- Click **“Setup”**
- Move the **Jig (3)** to the Position right under the R1 Probe.
- Select **“Adjust”**
- Perform Pulse adjustment that the R1 Probe touch the **Jig(3)**
- Click **“Accept”**

(5/6) R1 Washing Station (Outside Wash) (Vertical)

- Choose Adjustment position name **“(5/6) R1 Washing Station (Outside Wash) (Vertical)”**
- Click **“Setup”**
- Remove the **Jig (3)**
- Move the R Probe to the R Washing Station by hand.
- Select **“Adjust”**
- Perform Pulse adjustment that the height of the Washing Station matches the R1 probe tip.
- Click **“Accept”**

(6/6) R2 Washing Station (Outside Wash) (Vertical)

- Choose Adjustment position name **“(6/6) R2 Washing Station (Outside Wash) (Vertical)”**
- Click **“Setup”**
- Move the R Probe to the R Washing Station by hand.
- Select **“Adjust”**
- Perform Pulse adjustment so that the height of the Washing Station matches the R2 probe tip.
- Click **“Accept”** and **“End”**



3.16 Height Adjustment of R washing station position

Beads Mixing Position (Vertical)

- Place the **Jig (3)** in the Beads Mixing Position that the thick side is located to the BM side. (Figure 3.17)
- Select **“Beads Mixer”** from the Adjustment Selectionscreen.
- Chose Adjustment position name **“(3/3) Beads Mixing Position (Vertical)”**
- Click **“Setup”**
- Check that the paddle moves to the Beads Mixing Position.
- Select **“Adjust”**
- Perform Pulse adjustment so that the paddle touches the **Jig(3)**.
- Click **“Accept”** and **“End”**
- Remove the **Jig (3)**

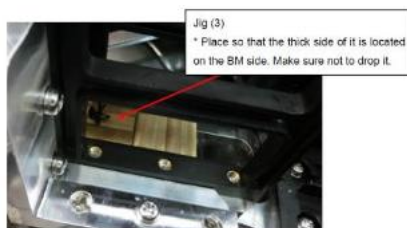
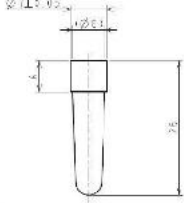
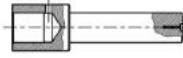
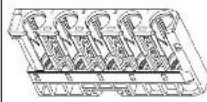


Figure 3.17 Mixing height

7.0 Sampling Adjustment Rotation

7.1 Adjustment Jigs

Name	AssayCup	Jig (1)	Jig (2)
Parts No.	7411801	7889260	
Item Name	AssayCup	Active Gripper Jig	
Quantity (pcs)	2	1	
Adjustment position	Inc position	SP position	SP position
Image or drawing			
Name	Standard cup	Rack	
Parts No.	-	-	
Item Name	-	-	
Quantity (pcs)	2	1	
Adjustment position	APC rack position	SP position	
Image or drawing			

7.2 Preparations

Remove following Covers:

Open the top cover (front/rear)

Rear Cover Assembly , PW Cover Assembly, Table Cover 7, Camera Protection Cover, Shielding Plate Assembly, INC Cover B, Table Cover 3, Table Cover 4

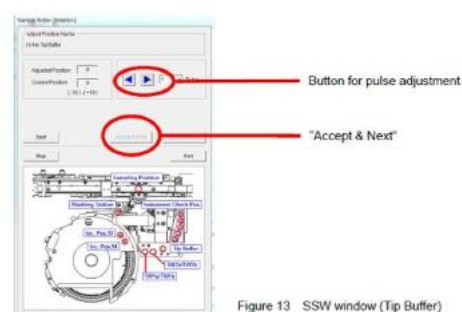
7.3 Software Adjustment

Preparation:

- Start SSW and select **"Service – Adjustment – Sample Probe (Rotation) (Figure 2)"**

(1/14) Tip Buffer

- Perform Pulse Adjustment that the S Nozzle is in the Center Tip Buffer ($\pm 0.3\text{mm}$) (Figure 13)
- Click **"Accept"** and **"Next"**



(2/14) Inc. Pos 94

- Place an **Assay Cup** in Position 94 of the Incubator
- Perform Pulse Adjustment that the S Nozzle is in the Center Assay Cup ($\pm 0.3\text{mm}$)
- Click "**Accept**" and "**Next**"

(3/14) Inc. Pos 51

- Place an **Assay Cup** in Position 51 of the Incubator
- Perform Pulse Adjustment that the S Nozzle is in the Center Assay Cup ($\pm 0.3\text{mm}$)
- Collect the Assay Cup
- Click "**Accept**" and "**Next**"

(4/14) TW 1B (Tip Waste)

- Perform Pulse Adjustment that the Part "a" of the S Nozzle is $1 \pm 0.5\text{mm}$ from the end face of TW 1B (Figure 14 and 15)
- Click "**Accept**" and "**Next**"

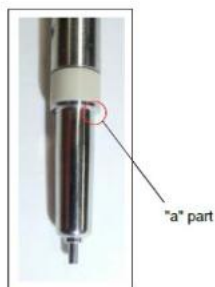


Figure 14 Sample probe

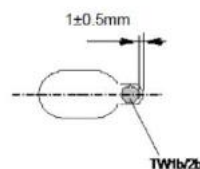


Figure 15 TW1b/2b adjustment position

(5/14) TW 2B (Tip Waste)

- Perform Pulse Adjustment that the Part "a" of the S Nozzle is $1 \pm 0.5\text{mm}$ from the end face of TW 2B (Figure 14 and 15)
- Click "**Accept**" and "**Next**"

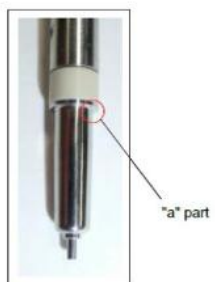


Figure 14 Sample probe

(6/14)

WP (Washing Position)

- Loosen S Washing Station fixing screws. (Figure 16)
- Adjust manually adjust the S Washing Station that the S Nozzle descends into the center.
- Click "**Next**"

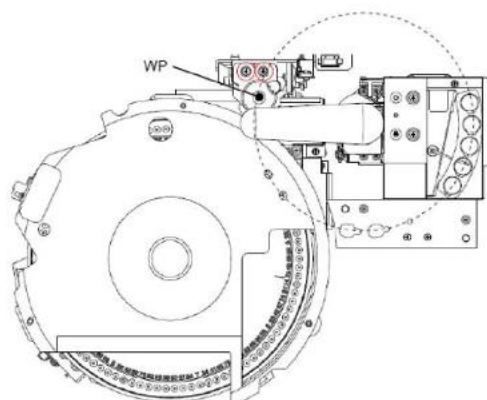


Figure 16 S washing station adjustment

(7/14) TW 1a

- Perform Pulse Adjustment that the Part "a" of the S Nozzle is $4.5 \pm 0.5\text{mm}$ from the end face of TW 1a (Figure 18)
- Click **"Accept"** and **"Next"**



Figure 14 Sample probe

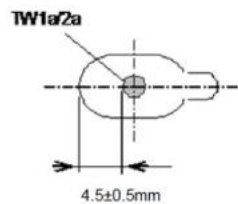


Figure 18 TW1a/2a adjustment position

(8/14) TW 2a

- Perform Pulse Adjustment that the Part "a" of the S Nozzle is $4.5 \pm 0.5\text{mm}$ from the end face of TW 1a (Figure 18)
- Click **"Accept"** and **"Next"**

(9/14) SP

- Place **Jig (2)** in Pos. 1 of the Rack and set the Rack into the Standby Position of L-Line (Figure 19)
- Click **"Sampling Position"** in the Sampling Rack Section of SSW
- Remove the Bridge Plate (3 screws) and unfasten the L-Line fixing screws (2 screws)
- Adjust the tip end of S Nozzle in the center of the hole of **Jig (2)** by using the tilt adjustment screw and the Pulse adjustment button in the SSW.
- After Adjustment, lift up the S Nozzle to the upper limit.
- Attach **Jig (1)** onto **Jig (2)**
- Make sure that the S Nozzle is in the center of the hole of **Jig(1)**

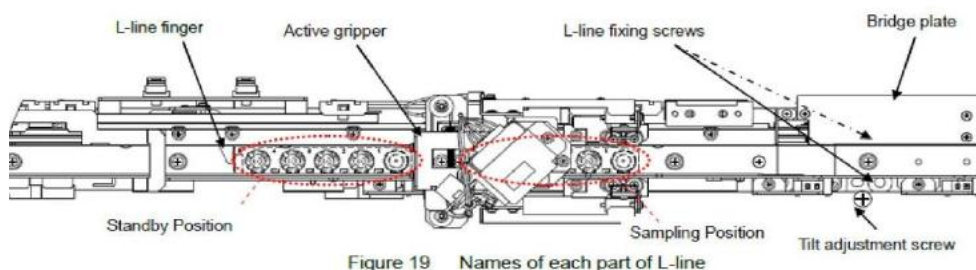


Figure 19 Names of each part of L-line

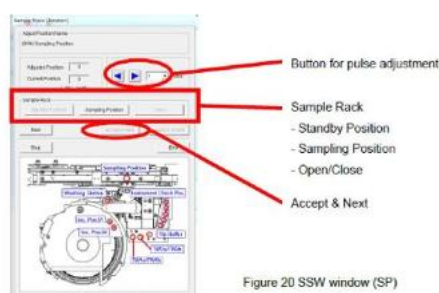


Figure 20 SSW window (SP)

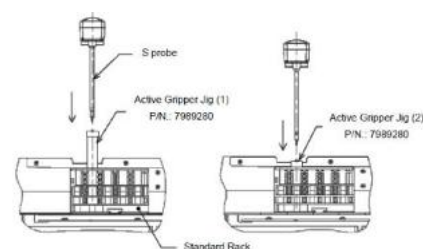


Figure 21 SP position adjustment

(10/14) Instrument Check Pos. 1

- Place a **standard cup** in the Banana Rack Pos. 1-5 (Figure 23)
- Perform Pulse Adjustment that the S Nozzle is in the Center of the Standard Cup ($\pm 0.3\text{mm}$)
- Click "**Accept**" and "**Next**"

(11/14) Instrument Check Pos. 2

- Perform Pulse Adjustment that the S Nozzle is in the Center of the Standard Cup ($\pm 0.3\text{mm}$)
- Click "**Accept**" and "**Next**"

(12/14) Instrument Check Pos. 3

- Perform Pulse Adjustment that the S Nozzle is in the Center of the Standard Cup ($\pm 0.3\text{mm}$)
- Click "**Accept**" and "**Next**"

(13/14) Instrument Check Pos. 4

- Perform Pulse Adjustment that the S Nozzle is in the Center of the Standard Cup ($\pm 0.3\text{mm}$)
- Click "**Accept**" and "**Next**"

(14/14) Instrument Check Pos. 5

- Perform Pulse Adjustment that the S Nozzle is in the Center of the Standard Cup ($\pm 0.3\text{mm}$)
- Click "**Accept**" and "**Next**"

Click "**End**" and remove all standard cups and Jigs.

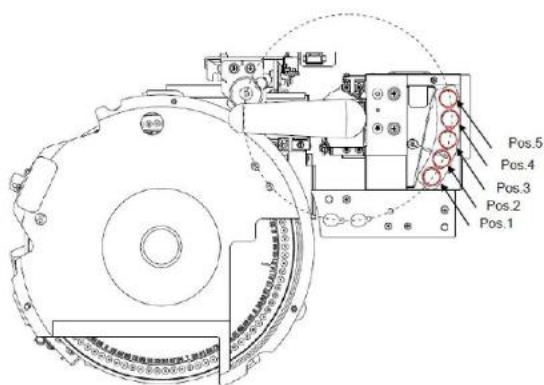
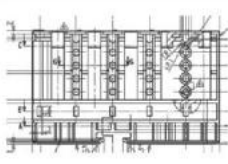


Figure 23 APC rack position adjustment

8.0 Sampling Adjustment Vertical

8.1 Adjustment Jigs

Name	AssayTip	Jig	Standard Rack
Parts No.		7676720	
Item Name	AssayTip	Sample pipetting Jig	
Quantity (pcs)	1	1	1
		SP position (height adjustment)	SP position (height adjustment)
Image or drawing			

8.2 Adjustment with SSW

Preparation:

- Start SSW and select “**Service – Adjustment – Sample Probe (Vertical) (Figure 1)**”
- Click “**Setup**”

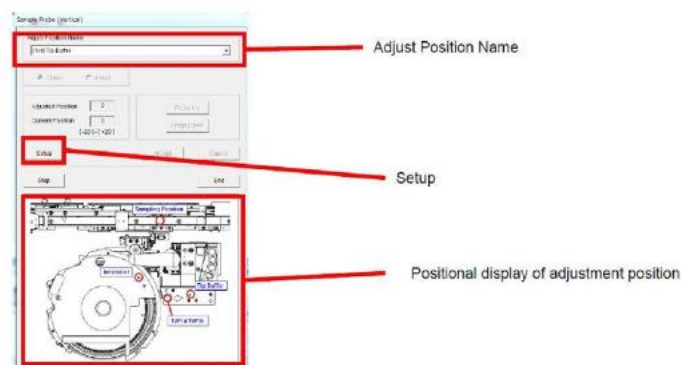


Figure 3 Initial window

(1/4) Tip Buffer

- Attach a Assay Tip to the Sampler Probe .
- Select “**(1/4) Tip Buffer**” in Adjusted Position Name
- Click “**Adjust**” (Figure 4)
- Perform Pulse Adjustment that the Assay Tip is in contact with the buffer plate.
- **Hint: To prevent a damage, choose 5 Steps in the upper Position, directly after choosing “Adjust”**
- Click “**Accept**”

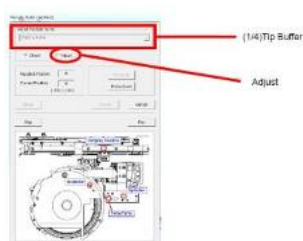


Figure 4 SSW window (Tip Buffer)

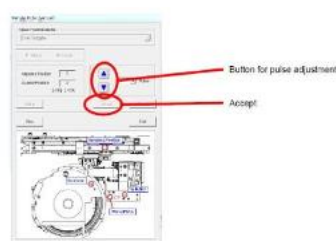
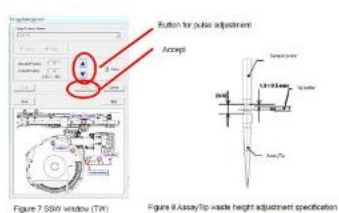
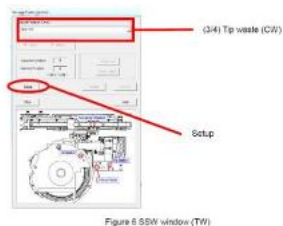


Figure 5 SSW window (Tip Buffer)

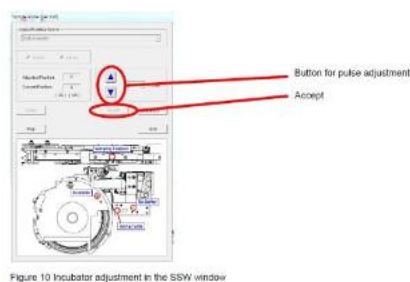
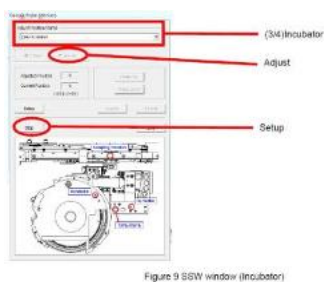
(2/4) TW- Tip Waste

- Select **“(2/4) TW”** in Adjusted Position Name and click **“Setup”**
- Select **“Adjust”** (Figure 6)
- Insert a 1.0mm spacer between the buffer plate and the probe neck.
- Perform Pulse Adjustment that they contact the spacer. (Figure 8)
- Reference Specification $1.3 \pm 0.5\text{mm}$
- Click **“Accept”**



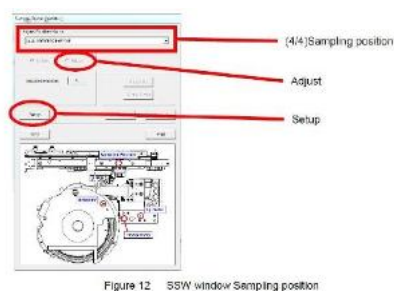
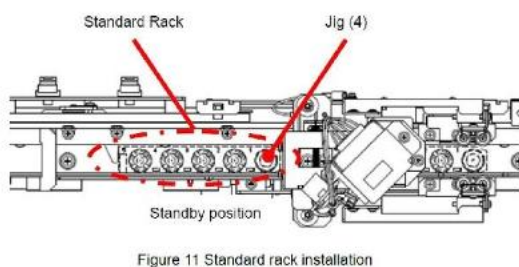
(3/4) Incubator

- Select **“(3/4) Incubator”** in Adjusted Position Name and click **“Setup”**
- Select **“Adjust”** (Figure 9)
- Perform Pulse Adjustment that the Assay Tip contacts the Incubator.
- Click **“Accept”** (Figure 10)



(4/4) Sampling Position

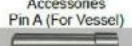
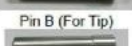
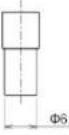
- Place **Jig (4)** on a Standard Rack and set it in the Standby Position of the L-Line. (Figure 11)
- Select **“(4/4) Sampling Position”** in Adjusted Position Name and click **“Setup”**
- Select **“Adjust”** (Figure 12)
- Make sure that the probe descends and the Assay Tip touch the **Jig (4)**.
- Collect the Rack and click **“End”**



9.0 T/V Gripper (X-Y Direction)

9.1 Adjustment Jigs

Table 1 List of jigs

Name	Jig (1)	Jig (2)	Jig (3)	Jig (4)	Jig (5)
Parts No.	798-7292	741-1471	741-8743	788-2562	768-3009
Name of item	Magazine Jig	Vessel Dummy Jig	Vessel Dummy Jig 3	Tip Jig	Dummy Vessel
Quantity(pcs)	1	1	1	1	1
Adjustment position	Magazine Position	Inc, Magazine (Vessel), Mixer	Magazine	Magazine (Tip), Tip Buffer	Magazine VM VW
Image or drawing	 Accessories  Pin A (For Vessel)  Pin B (For Tip)	 $\Phi 6.1$			

9.2 Preparations

Remove following Covers:

TV Cover (BL) assembly, R Cover, Incubator Cover B, TV Cover (Z), Sensor Cover (X and Y Direction)

9.3 Changed after SSW 06-01

In SSW 06-01 , the Drop Down menu section has been removed for the X-Y adjustment . Adjustments will start from position “(1/16) Home Position” and will follow to next position by pressing “Next” or “Accept & Next” button .. All the movements between the positions will be done automatically .

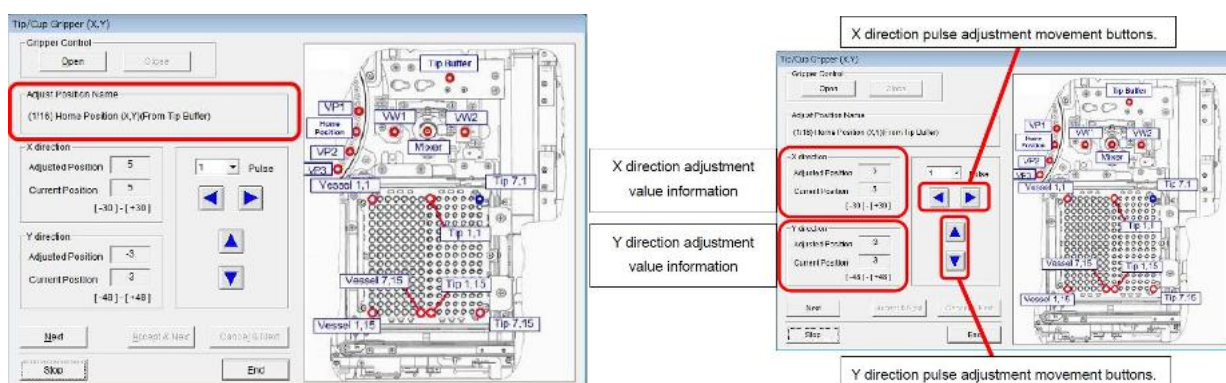
9.4 Adjustment with SSW

Preparation

- Place the **Jig (1)** on the magazine fixer. Is there a magazine, remove it and place the **Jig (1)**
- Select “ **Service – Adjustment – Tip/Cup Gripper (X,Y)** ” and click “**Adjust**”

(1/16) Home Position (X,Y) (From Tip Buffer)

- Make sure that “ Adjust Position Name ” is “ **(1/16) Home Position (X,Y) (From Tip Buffer)** ” as shown in the Figure below.
- Place **Jig (2)** to the Gripper
- Move the Gripper Downward into the hole on Incubator Position 1
- Perform Pulse Adjustment in X and Y Direction that the **Jig (2)** is in the center of the hole
- Click “**Accept & Next**”



(2/16) Home Position (Y) (From Magazine)

- Move the Gripper downward into the hole on Incubator (Home Position)
- Perform Pulse Adjustment in Y Direction that the **Jig (2)** is in the center of the Home Position
- Click “**Accept & Next**”
- Remove the **Jig (2)**

(3/16) Vessel 1, 15

- Place **Jig (3)** to the Gripper
- Move the Gripper downward into the hole on Vessel 1, 15 Position
- Perform Pulse Adjustment in X and Y Direction that the **Jig (3)** is in the center of Vessel 1, 15 Position
- Click “**Accept & Next**”

(4/16) Vessel 1, 1

- Move the Gripper with the **Jig (3)** downward into the hole on Vessel 1, 1 Position
- Perform Pulse Adjustment in X Direction that the **Jig (3)** is in the center of Vessel 1, 1 Position
- Click “**Accept & Next**”

(5/16) Vessel 7, 15

- Move the Gripper with the **Jig (3)** downward into the hole on Vessel 7, 15 Position
- Perform Pulse Adjustment in Y Direction that the **Jig (3)** is in the center of Vessel 7, 15 Position
- Click **"Accept & Next"**
- Remove **Jig (3)**

(6/16) Tip 1, 15

- Place **Jig (4)** to the Gripper
- Move the Gripper with the **Jig (4)** downward into the hole on Tip 1, 1 Position
- Perform Pulse Adjustment in X and Y Direction that the **Jig (4)** is in the center of Tip 1, 15 Position
- Click **"Accept & Next"**

(7/16) Tip 1, 1

- Move the Gripper with the **Jig (4)** downward into the hole on Tip 1, 1 Position
- Perform Pulse Adjustment in X and Y Direction that the **Jig (4)** is in the center of Tip 1, 15 Position
- Click **"Accept & Next"**

(8/16) Tip 7, 15

- Move the Gripper with the **Jig (4)** downward into the hole on Tip 7, 15 Position
- Perform Pulse Adjustment in Y Direction that the **Jig (4)** is in the center of Tip 7, 15 Position
- Click **"Accept & Next"**

(9/16) Tip 7, 1 (Control Position)

- Move the Gripper with the **Jig (4)** downward into the hole on Tip 7, 1 Position
- Check the Adjustment in Y Direction that the **Jig (4)** is in the center of Tip 7, 1 Position
- Click **"Next"**

(10/16) Tip Buffer

- Move the Gripper with the **Jig (4)** downward into the hole of the Tip Buffer
- Perform Pulse Adjustment in x and Y Direction that the **Jig (4)** is in the center of Tip Buffer
- Click **"Accept & Next"**
- Remove **Jig (4)**

(11/16) Mixer

- Place **Jig (2)** to the Gripper
- Move the Gripper downward into the hole of the Mixer Position
- Perform Pulse Adjustment in X Direction that the **Jig (2)** is in the center of the Mixer
- Click **“Accept & Next”**
- Remove **Jig (2)**

(12/16) VW1

- Place **Jig (5)** to the Gripper
- Move the Gripper downward into the hole of VW1
- Perform Pulse Adjustment in X Direction that the **Jig (5)** is in the center of the VW1
- For Y Direction Hardware Adjustment is needed.
- Click **“Accept & Next”**

(13/16) VW2

- Move the Gripper downward into the hole of VW2
- Perform Pulse Adjustment in X Direction that the **Jig (5)** is in the center of the VW2
- For Y Direction Hardware Adjustment is needed.
- Click **“Accept & Next”**
- Remove **Jig (5)**

(14/16) VP2

- Place **Jig (2)** to the Gripper
- Move the Gripper downward into the hole of the Incubator Position 44
- Perform Pulse Adjustment in X and Y Direction that the **Jig (2)** is in the center of VP2
- Click **“Accept & Next”**

(15/16) VP3

- Move the Gripper downward into the hole of the Incubator Position 87
- Perform Pulse Adjustment in X and Y Direction that the **Jig (2)** is in the center of VP3
- Click **“Accept & Next”**

(16/16) VP1

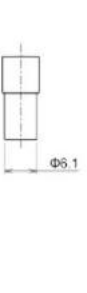
- Move the Gripper downward into the hole of the Incubator Position 52
- Perform Pulse Adjustment in X and Y Direction that the **Jig (2)** is in the center of VP1
- Click **“Accept & Next”**
- Remove **Jig (2)**

When finishing the Software X-Y Direction Adjustment click the **“End”** button.

10.0 T/V Gripper (Z- Direction)

10.1 Adjustment Jigs

Table 1 List of jigs

Name	Jig (1)	Jig (2)	Jig (3)	Jig (4)	Jig (5)
Parts No.	798-7292	741-1471	741-8743	788-2562	768-3009
Name of item	Magazine Jig	Vessel Dummy Jig	Vessel Dummy Jig 3	Tip Jig	Dummy Vessel
Quantity(pcs)	1	1	1	1	1
Adjustment position	Magazine Position	Inc. Magazine (Vessel), Mixer	Magazine	Magazine (Tip), Tip Buffer	Magazine VM VW
Image or drawing					

10.2 Preparations

Remove following Covers:

TV Cover (BL) assembly, R Cover, Incubator Cover B, TV Cover (Z), Sensor Cover (X and Y Direction)

10.3 Adjustment with SSW

Preparation

- Place the **Jig (1)** on the magazine fixer. Is there a magazine, remove it and place the **Jig (1)**
- Select **"Service – Adjustment – Tip/Cup Gripper (Vertical)"** and click **"Adjust"**

(1/12) Incubator

- Select **"(1/12) Incubator"** from Adjust Position Name and click **"Setup"** (Figure 3-7)
- Place **Jig (5)** in Incubator Position 1
- Click **"Open"**
- Click **"Gripper Down"** and **"Close"**
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface of the Jig (5)
- **"Open – Up / Down – Close"** and check
- Click **"Accept"**
- Remove the **Jig (5)**

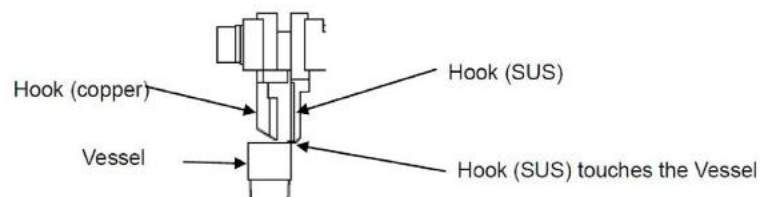


Figure 4-2 Adjustment position of the gripper height.

(5/12) Vessel 1, 1

- Select “ **(5/12) Vessel 1, 1**” and click “**Setup**”
- Insert **Pin A (from Jig (1))** into Vessel 1, 1 Position
- Click “**Open**”
- Click “**Gripper Down**” and “**Close**”
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface (Figure 4.3)
- “**Open – Up / Down – Close**” and check
- Click “**Accept**”

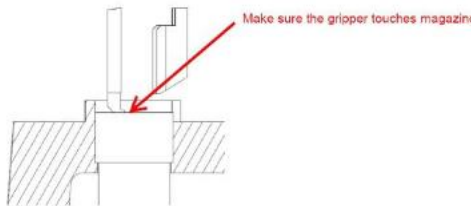


Figure 4-3 Adjustment position of gripper on magazine

(6/12) Vessel 1, 15

- Select “ **(6/12) Vessel 1, 15**” and click “**Setup**”
- Insert **Pin A (from Jig (1))** into Vessel 1, 15 Position
- Click “**Open**”
- Click “**Gripper Down**” and “**Close**”
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface (Figure 4.3)
- “**Open – Up / Down – Close**” and check
- Click “**Accept**”

(11/12) Tip 7, 1

- Select “ **(11/12) Tip 7, 1**” and click “**Setup**”
- Insert **Pin B (from Jig (1))** into Tip 7, 1 Position
- Click “**Open**”
- Click “**Gripper Down**” and “**Close**”
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface (Figure 4.3)
- “**Open – Up / Down – Close**” and check
- Click “**Accept**”

(12/12) Tip 7, 15

- Select “ **(12/12) Tip 7, 15**” and click “**Setup**”
- Insert **Pin B (from Jig (1))** into Tip 7, 15 Position
- Click “**Open**”
- Click “**Gripper Down**” and “**Close**”
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface (Figure 4.3)
- “**Open – Up / Down – Close**” and check
- Click “**Accept**”

(2/12) Tip Buffer

- Select “ **(2/12) Tip Buffer**” and click “**Setup**”
- Place **Jig (4)** in the Tip Buffer Position
- Click “**Open**”
- Click “**Gripper Down**” and “**Close**”
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface (Figure 4.2)
- “**Open – Up / Down – Close**” and check
- Click “**Accept**”, “**Setup**” and remove the **Jig (4)**

(3/12) Mixer

- Select “ **(3/12) Mixer**” and click “**Setup**”
- Place **Jig (5)** in the Mixer Position
- Click “**Open**”
- Click “**Gripper Down**” and “**Close**”
- Perform Pulse Adjustment that the Gripper Hook (SUS) is touching the surface (Figure 4.2)
- “**Open – Up / Down – Close**” and check
- Click “**Accept**”, “**Setup**” and remove the **Jig (5)**

(4/12) VW

- Select “ **(4/12) VW**” and click “**Setup**”
- Click “**Gripper Down**”
- Click “**Open**”
- Perform Pulse Adjustment that the Gripper Hook is 1.0 ~ 2.0 mm above the surface (Figure 4.4)
- Click “**Close**”
- Click “**Accept**”

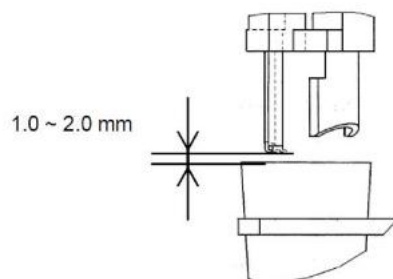
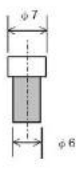
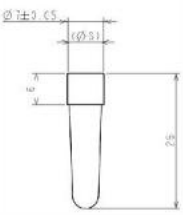


Figure 4-4 Height adjustment position of Vessel discarding pipe

When finishing the Vertical Adjustment, click “**End**”

11.0 Detection Unit

11.1 Adjustment Jigs

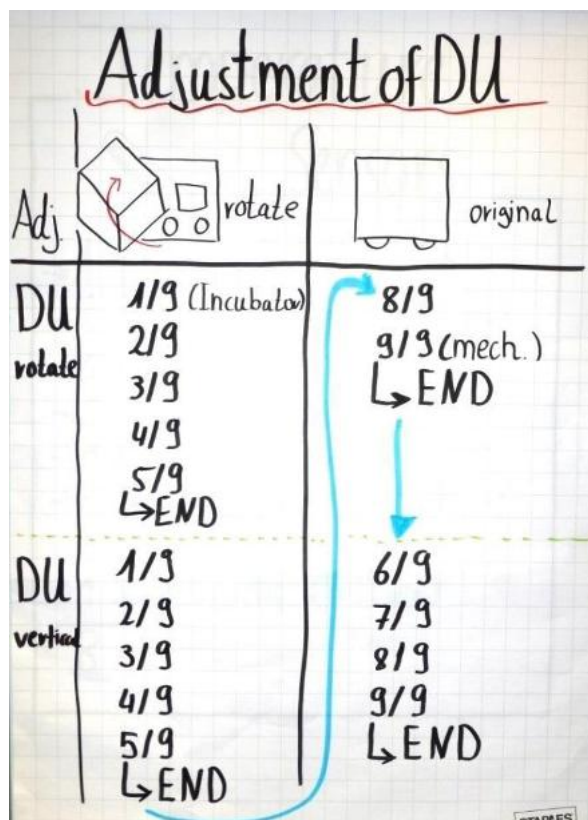
Name	Jig (1)	Jig (2)	Jig (3)	Jig (4)
Parts No.	(P/N:741-1471)	(P/N:741-1801)	(P/N:786-9795)	(P/N:786-3304)
Name	Vessel Dummy Jig	AssayCup	Sipper Nozzle Jig	BMRS Jig
Quantity (pcs)	1	3	1	1
Adjust position			Reaction mixture aspiration position	Height of washing station
Image or drawing				

11.2 Preparation

Remove following Covers:

Incubator Cover B, Table Cover (Left/right side), R jacket Cover 2, DU Side Cover, DU Front Cover, DU Duct Cover 1, DU Duct Cover 2, DU Cover Lid (attention for the clamp of O-tubes)

11.3 Adjustment Flowchart



11.4 Software Adjustment

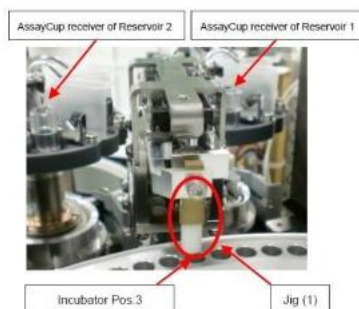
Rotation (DU is turned)

Preparation:

- Turn DU to the side
- Start SSW and select **“Service – Adjustment – Detection Rotation”** then click **“Adjust”**

(1/9) Home -> Incubator

- Attach **Jig (1)** to the Sipper Gripper
- Move the Sipper Gripper downwards manually
- Perform Pulse Adjustment that the **Jig (1)** can be inserted to the hole in the Incubator
- Click **“Accept”** and **“Next”**



(2/9) Incubator -> Reservoir 2

- Move the Sipper Gripper downwards manually
- Perform Pulse Adjustment that the **Jig (1)** can be inserted into the Assay Cup receiver of Reservoir 2
- Click **“Accept”** and **“Next”**



(3/9) Reservoir 2 -> Incubator

- Move the Sipper Gripper downwards manually
- Perform Pulse Adjustment that the **Jig (1)** can be inserted into the Assay Cup receiver of Incubator.
- Click **“Accept”** and **“Next”**
- Remove the **Jig (1)**

- Click **“End”** when height adjustment is not performed.

(4/9) Incubator -> Reservoir 1

- Move the Sipper Gripper downwards manually
- Perform Pulse Adjustment that the **Jig (1)** can be inserted into the Assay Cup receiver of Reservoir 1
- Click **“Accept”** and **“Next”**

(5/9) Reservoir -> 1 Incubator

When a height adjustment is not performed, choose **End** in **(5/8) Reservoir1 Incubator**.

Height Adjustment (DU is turned):

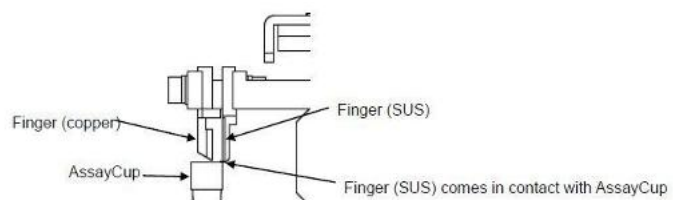
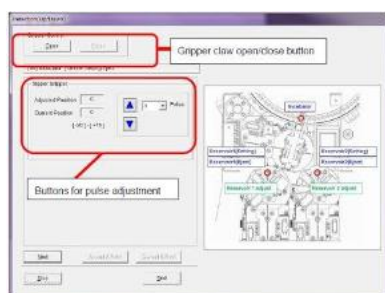
Preparation:

- Start SSW and select **“Service – Adjustment – Detection (Vertical)”** then click **“Adjust”**
- Place **Assay Cups** in Incubator Position 3, Reservoir 1 and 2



(1/9) Incubator

- Item **“(1/9) Incubator”** appears
- Click **“Open”**
- Perform Pulse Adjustment that the lower edge of the finger (SUS) touches the **AssayCup** top surface by reducing 1 Pulse
- **“Open – Up / Down – Close”** and check
- The adjustment position is 1 Pulse above this point
- Click **“Accept”** and **“Next”**



(2/9) Reservoir 1 (Loading)

- Item **“(2/9) Reservoir 1”** appears
- Click **“Open”**
- Perform Pulse Adjustment that the lower edge of the finger (SUS) touches the **AssayCup** top surface by reducing 1 Pulse
- **“Open – Up / Down – Close”** and check
- Click **“Accept”** and **“Next”**

(3/9) Reservoir 1 (unloading)

- Item **“(3/9) Reservoir 1”** appears
- Click **“Open”**
- Perform Pulse Adjustment that the lower edge of the finger (SUS) touches the **AssayCup** top surface by reducing 1 Pulse
- **“Open – Up / Down – Close”** and check
- Click **“Accept”** and **“Next”**

(4/9) Reservoir 2(Loading)

- Item **“(4/9) Reservoir 1”** appears
- Click **“Open”**
- Perform Pulse Adjustment that the lower edge of the finger (SUS) touches the **AssayCup** top surface by reducing 1 Pulse
- **“Open – Up / Down – Close”** and check
- Click **“Accept”** and **“Next”**

(5/9) Reservoir 2 (unloading)

- Item **“(5/9) Reservoir 2”** appears
- Click **“Open”**
- Perform Pulse Adjustment that the lower edge of the finger (SUS) touches the **AssayCup** top surface by reducing 1 Pulse
- **“Open – Up / Down – Close”** and check
- Click **“Accept”** and **“Next”**
- Click **“End”** and remove the **Jigs (Assay Cups)**

Return the DU to the original Position

- Make sure that the Reservoirs are in the lower Position
- Be careful not to disconnect the wires on the rear side of DU

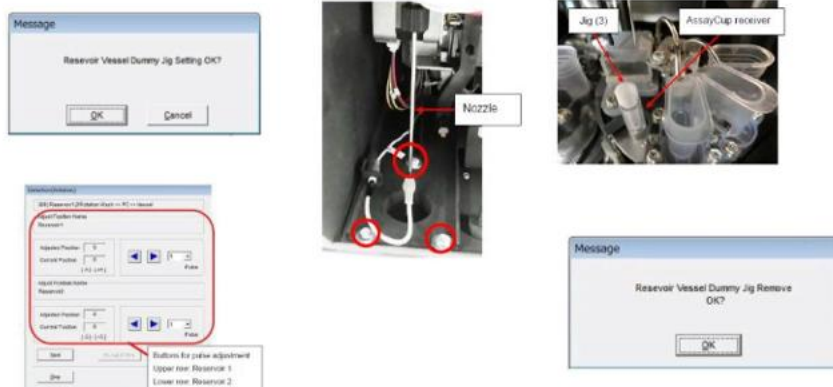
Continue of DU Rotation Adjustment:

Preparation:

- Turn DU to the original Position
- Start SSW and select **“Service – Adjustment – Detection Rotation”** then click **“Adjust”**
- Click **“Next”** until **“(8/9) Reservoir -> Rotation Wash -> PC -> Wash”** appears

(8/9) Reservoir -> Rotation Wash -> PC -> Wash

- Place **Jig (3)** on the Reservoir and click **"OK"**
- Lift up the Reservoir manually.
- Perform Pulse Adjustment that the Sipper Nozzle is perfectly in the center of **Jig (3)**
- Individual fine Adjustment can be done manually with the fixing screws of the sipper nozzles (Figure 3.20)
- Click **"Accept"** and **"Next"** and **"End"**
- A Window to confirm to remove the **Jig (3)** will appear. Click **"OK"**



Continue of DU Vertical Adjustment:

Preparation:

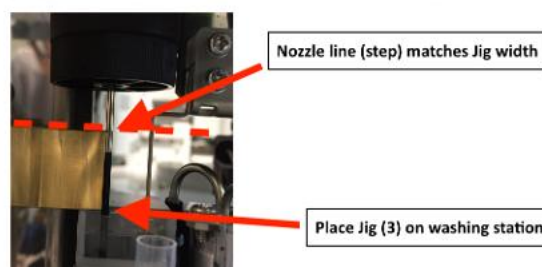
- Select **"Service – Adjustment – Detection Vertical"** then click **"Adjust"**
- Click **"Next"** until **"(6/9) Reservoir 1: Aspiration"** appears
- Place **Assay Cups** in Reservoir 1 and 2

(6/9) Reservoir 1: Aspiration

- Item **"(6/9) Reservoir 1: Aspiration"** appears
- Click **"Open"**
- Perform Pulse Adjustment that the nozzle tip contacts the AssayCup bottom and the cup is unmovable upward and downward
- Click **"Accept"** and **"Next"**

(7/9) Reservoir 1: Wash

- Item **"(6/9) Reservoir 1: Wash"** appears
- Click **"Open"**
- Perform pulse adjustment so that the height from the top of the washing station to the line in the upper part of the Nozzle (step) is equal to the width of the **Jig(4)**
- Click **"Accept"** and **"Next"**



(8/9) Reservoir 2: Aspiration

- Item **“(8/9) Reservoir 2: Aspiration”** appears
- Click **“Open”**
- Perform Pulse Adjustment that the nozzle tip contacts the Assay Cup bottom and the cup is unmovable upward and downward
- Click **“Accept”** and **“Next”**

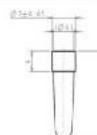
(9/9) Reservoir 2: Wash

- Item **“(9/9) Reservoir 2: Wash”** appears
- Click **“Open”**
- Perform pulse adjustment so that the height from the top of the washing station to the line in the upper part of the Nozzle (step) is equal to the width of the **Jig (4)**
- Click **“Accept”** and **“Next”** and **“End”**

A Window to confirm to **remove the Jigs** will appear. Click **“OK”**

12.0 Prewash Gripper

12.1 Adjustment Jigs

Name	Jig (1)	Jig (2)	Jig (3)	Jig (4)	Jig (5)
Parts No.	7411471	7889688	Consumable part	7980367	7683009
Item Name	Vessel Dummy Jig	PW sipper Jig	AssayCup	PW-LFC cup assembly (accessory part)	Dummy Vessel
Quantity (pcs)	1	1	5	1	1
Adjustment position	Inc position	Washing station position	Gripper height	LFC position	SS position
Image or drawing					

12.2 Preparations

Remove following Covers:

Remove the rear Cover, INC Cover B, PreWash Cover, Table Cover 5, Table Cover 4

12.3 Adjustment with SSW

Preparation

- Select **“Service – Adjustment – PreWash Gripper”** and click **“Adjust”**

Horizontal Adjustment

(1/10) Home Position (X-Direction)

- Make the pre-wash gripper grip the **Jig(1)** with its claw
- Move the Gripper downward into the hole on Incubator Position 7
- Perform Pulse Adjustment that the **Jig (1)** is in the center of the Incubator Position 7
- Click **“Accept”**

(2/10) Mixer 1 (X-Direction)

- Select **“(2/10) Mixer1 (X-Direction)”** in Adjustment Position Name, then click **“Setup”**
- Move the Gripper downward into the Mixer 1
- Perform Pulse Adjustment that the **Jig (1)** is in the center of the Mixer 1
- Click **“Accept”**

(3/10) Mixer 2 (X-Direction)

- Select **“(3/10) Mixer 2 (X-Direction)”**, then click **“Setup”**
- Move the Gripper downward into the Mixer 2

- Perform Pulse Adjustment that the **Jig (1)** is in the center of the Mixer 2
- Click **"Accept"**

(4/10) SS1 (X-Direction)

- Select **"(4/10) Mixer1 (X-Direction)"**, then click **"Setup"**
- Move the Gripper downward into the SS1 Position
- Perform Pulse Adjustment that the **Jig (1)** is in the center of the SS1
- Click **"Accept"**

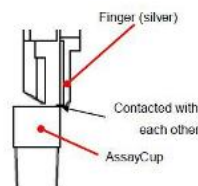
(5/10) SS2 (X-Direction)

- Select **"(5/10) Mixer1 (X-Direction)"**, then click **"Setup"**
- Move the Gripper downward into the SS1 Position
- Perform Pulse Adjustment that the **Jig (1)** is in the center of the SS2
- Click **"Accept"** and **remove Jig(1)**

Vertical Adjustment

(6/10) Home Position (Z-Direction)

- Place the **Jig(3) (AssayCup)** in the incubator position 7.
- Select **"(6/10) Home Position (Z-Direction)"** in Adjustment Position Name, then click **"Setup"**
- Click **"Open"**
- Click **"Gripper Down"** and **"Close"**
- Perform Pulse Adjustment until the claw of the pre-wash gripper contacts the top of **Jig(3)**
- Make sure that the gripper fingers have been closed.
- Click **"Accept"**



(7/10) Mixer 1 (Z-Direction)

- Select **"(6/10) Home Position (Z-Direction)"**, then click **"Setup"**
- Place the **Jig(3) (AssayCup)** in the Mixer Position 1
- Select **"(7/10) Mixer 1 (Z-Direction)"**, then click **"Setup"**
- Click **"Open"**
- Click **"Gripper Down"** and **"Close"**
- Perform Pulse Adjustment until the claw of the pre-wash gripper contacts the top of **Jig(3)**
- Make sure that the gripper fingers have been closed.
- Click **"Accept"**

(9/10) SS 1 (Z-Direction)

- Select **“(6/10) Home Position (Z-Direction)”**, then click **“Setup”**
- Place the **Jig(3) (AssayCup)** in the SS 1 Position
- Select **“(9/10) SS 1 (Z-Direction)”**, then click **“Setup”**
- Click **“Open”**
- Click **“Gripper Down”** and **“Close”**
- Perform Pulse Adjustment until the claw of the pre-wash gripper contacts the top of **Jig(3)**
- Make sure that the gripper fingers have been closed.
- Click **“Accept”**

(10/10) SS 2 (Z-Direction)

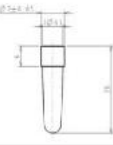
- Select **“(6/10) Home Position (Z-Direction)”** in **Adjustment Position Name**, then click **“Setup”**
- Place the Jig(3) (AssayCup) in the SS 2 Position
- Select **“(10/10) SS 2 (Z-Direction)”** in **Adjustment Position Name**, then click **“Setup”**
- Click **“Open”**
- Click **“Gripper Down”** and **“Close”**
- Perform Pulse Adjustment until the claw of the pre-wash gripper contacts the top of **Jig(3)**
- Make sure that the gripper fingers have been closed
- Click **“Accept”**

Finish PreWash Gripper Adjustment

- Select **“(6/10) Home Position (Z-Direction)”**, then click **“Setup”**
- Click **“End”** to complete the PreWash Gripper Adjustment
- **Remove all Jigs**

13.0 Prewash Sipper

13.1 Adjustment Jigs

Name	Jig (1)	Jig (2)	Jig (3)	Jig (4)	Jig (5)
Parts No.	7411471	7889888	Consumable part	7980367	7683009
Item Name	Vessel Dummy Jig	PW sipper Jig	AssayCup	PW-LFC cup assembly (accessory part)	Dummy Vessel
Quantity (pcs)	1	1	5	1	1
Adjustment position	Inc position	Washing station position	Gripper height	LFC position	SS position
Image or drawing					

13.2 Preparations

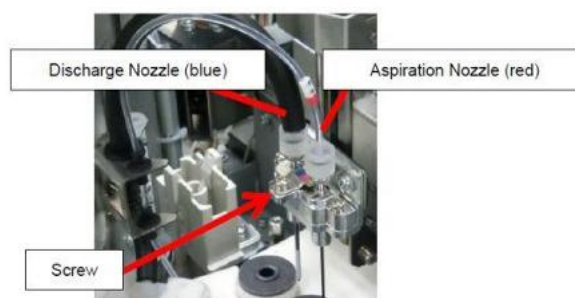
Remove following Covers:

Remove the rear Cover, INC Cover B, PreWash Cover, Table Cover 5, Table Cover 4

13.3 Adjustment with SSW

Preparation

- Select **“Service – Adjustment – PreWash Sipper”** and click **“Adjust”**



Horizontal Adjustment

(1/12) Home Position (X-Direction)

- Select **“(1/12) Home Position (X-Direction)”** in Adjustment Position Name, then click **“Setup”**
- Set **Jig (2)** on the Washing Station
- Perform Pulse Adjustment that the **pre-wash sipper nozzle (red)** will be inserted into the hole of the **Jig(2)**
- Click **“Accept”**

(2/12) Washing Liquid Discard (X-Direction)

- Select **“(2/12) Washing Liquid Discard (X-Direction)”**, then click **”Setup”**
- Perform Pulse Adjustment that the **pre-wash sipper nozzle (blue)** will be inserted into the hole of the **Jig(2)**
- Click **”Accept”**

(3/12) SS1 Aspiration (X-Direction)

- Select **“(3/12) SS1 Aspiration (X-Direction)”**, then click **”Setup”**
- Set **Jig (3)** on SS1 and SS2
- Perform Pulse Adjustment that the **pre-wash sipper nozzle (red)** so that it will be near the center of the **Jig(3)**
- Click **”Accept”**

(4/12) SS1 Discharging (X-Direction)

- Select **“(4/12) SS1 Discharging (X-Direction)”**, then click **”Setup”**
- Perform Pulse Adjustment that the **pre-wash sipper nozzle (blue)** so that it will be near the center of the **Jig(3)**
- Click **”Accept”**

(5/12) SS2 Aspiration (X-Direction)

- Select **“(5/12) SS2 Aspiration (X-Direction)”**, then click **”Setup”**
- Perform Pulse Adjustment that the **pre-wash sipper nozzle (red)** so that it will be near the center of the **Jig(3)**
- Click **”Accept”**

(6/12) SS2 Discharging (X-Direction)

- Select **“(6/12) SS2 Discharging (X-Direction)”**, then click **”Setup”**
- Perform Pulse Adjustment that the **pre-wash sipper nozzle (blue)** so that it will be near the center of the **Jig(3)**
- Click **”Accept”**

(7/12) LFC (X-Direction)

- Select **“(1/12) Home Position (X-Direction)”**, then click **”Setup”**
- Attach the **Table Cover 5** and set the **Jig(4)**
- Select **“(7/12) LFC (X-Direction)”**, then click **”Setup”**
- Perform Pulse Adjustment that the pre-wash sipper nozzle position until the tip end is inserted into the nozzle access hole of the LFC cup
- Click **”Accept”**

Vertical Adjustment

(8/12) SS1 Aspiration (Z-Direction)

- Place **Jig (5)** on SS1 and SS2
- Select **“(8/12) SS1 Aspiration (Z-Direction)”** in Adjustment Position Name, then click **”Setup”**
- Select **”Adjust”**
- Perform Pulse Adjustment until the height of the pre-wash sipper nozzle contacts the **Jig(5)**
- Click **”Accept”**

(9/12) SS1 Discarding (Z-Direction)

- Select **“(9/12) SS1 Discarding (Z-Direction)”**, then click **”Setup”**
- Select **”Adjust”**
- Perform Pulse Adjustment until the height of the pre-wash sipper nozzle contacts the **Jig(5)**
- Click **”Accept”**

(10/12) SS2 Aspiration (Z-Direction)

- Select **“(10/12) SS2 Aspiration (Z-Direction)”**, then click **”Setup”**
- Select **”Adjust”**
- Perform Pulse Adjustment until the height of the pre-wash sipper nozzle contacts the **Jig(5)**
- Click **”Accept”**

(11/12) SS2 Discarding (Z-Direction)

- Select **“(11/12) SS2 Discarding (Z-Direction)”**, then click **”Setup”**
- Select **”Adjust”**
- Perform Pulse Adjustment until the height of the pre-wash sipper nozzle contacts the **Jig(5)**
- Click **”Accept”**

(12/12) LFC (Z-Direction)

- Select **“(12/12) Home Position (X-Direction)”**, then click **”Setup”**
- Place the **Jig(4)**
- Select **”Adjust”** and the nozzles descent to the current adjustment position, then push manually the PW unit in X-direction so that the dispense nozzle is placed over the grey plastic frame.
- Perform Pulse Adjustment until the tip end of the **pre-wash sipper nozzle (red)** touches the surface of the **Jig(4)**. Make sure the nozzle does not tilt the plastic frame.
- Click **”Accept”**
- Select **“(1/12) Home Position (X-Direction)”** in Adjustment Position Name, then click **”Setup”**
- **Remove the Jigs**
- Click **”End”** to complete the Adjustment

